

		All in USD				
	Count	Cost Per Unit	Total Cost (before taxes & shipping)	Name	Link	Notes
ELECTRONICS	10	3.16	31.6	TowerPro Servo Motor MG996R 180	https://hacktronics.co/product/mg996r-torque-digital-metal-gear-servo-for-model-helicopter-rc-car-boat/	Servos used in knee and hip joints.
	1	110	110	Raspberry Pi 5 (4GB Ram)	https://www.pishop.us/product/raspberry-pi-5-4gb/?src=raspberrypi	Main computer of MIST. Compute allows us to run ollama for autonomous vision system.
	4	14.99	59.96	D2822 Fixed Wing Motor	https://www.amazon.com/D2822-Fixed-Wing-MultiCopter-Quadcopter-Multi-axis/dp/B0D3DYJSSL/	The motors powering forward acceleration.
	1	24.99	24.99	Power Distribution Board 2 (8 connectors)	https://www.servocity.com/xt30-power-distribution-board-xt90-input-8-x-xt30-outputs/	PDB necessary to distribute power without having to hotwire every connection.
	4	8.77	35.08	Bidirectional Brushless ESC 30A	https://www.aliexpress.com/i/3256807944824104.html?gatewayAdapt=4itemAdapt	Motor controller for the D2822 motors powering forward acceleration.
	1	35	35	Hosyond 5 Inch Touchscreen IPS MIPI DSI Display	https://www.amazon.com/Hosyond-Touchscreen-Compatible-Capacitive-Driver-Free/dp/B0CXTFN8K9/	Mist's screen and face.
	1	30	30	Turnigy 5000mAh 3S 30C 11.1V Lipo Battery Pack w/ XT90	https://speedyfpv.com/products/turnigy-5000mah-3s-30c-lipo-pack-w-xt-90?variant=48194865922262	Lipo battery powering all components of MIST. Any 11.1V Lipo battery will work. mAh is just a measurement of how long the battery will last. Higher mAh is equal to longer battery life.
	1	16.64	16.64	5MP 1080P Optional Night Vision Wide Angle IR-CUT FF AF Camera	https://www.aliexpress.us/item/3256806603685338.html	Camera used for vision and automation.
	2	12	24	Power Distribution Board 1 (6 connectors)	https://www.aliexpress.us/item/3256809728763176.html?	PDB necessary to distribute power without having to hotwire every connection.
	1	10	10	2pcs BEC UBEC 5V 5A	https://www.amazon.com/Module-Quadcopter-Airplane-Servo-Model/dp/B0D97FH1JS/	You need 2 UBECs. This offering provides both.
	4	2.5	10	TowerPro MG90S Servo Motor	https://www.tinyosshop.com/index.php?route=product/product&product_id=84	Servos used in arm.
	1	9.36	9.36	6mm Precision Mini Micro Planetary Reduction Reducer Gear Motor	https://www.amazon.com/Precision-Planetary-Reduction-Consistency-Torsion/dp/B08VRYVYBS/	Compact DC motor that fits and is used in gripper.
	1	19.99	19.99	Logitech - F310 Controller	https://www.bestbuy.com/product/logitech-f310-gaming-pad-for-windows-black-blue/J7H7ZLVG2X	You can choose another controller if you'd like, if it doesn't work upon initialization just ensure you modify the code to make it compatible.
	2	4.86	9.72	PCA9685PW	https://www.aliexpress.us/item/3256806112519208.html?	Generic PCA9685 brand. Any other model will work too (including the Adafruit model).
	1	3	3	MPU-6050	https://www.adafruit.com/product/3886	Accelerometer used to determine MIST's position and orientation and output back to the Raspberry Pi.

MATERIALS	1	19.97	19.97	SEBS Filament for wheel sleeves	https://hartsmartproducts.com/products/jabil-tpu95a-sebs-1300	Flexible filament used in wheel covers.
	1	19.99	19.99	White PLA Basic Filament (1kg) Bambu Lab US Store	https://us.store.bambulab.com/products/pla-basic-filament?id=41078274687112	Filament used in parts across MIST.
	1	19.99	19.99	Grey PLA Basic (1kg) Bambu Lab US Store	https://us.store.bambulab.com/products/pla-basic-filament?id=42844628975752	Filament used in parts across MIST.
	1	17.72	19.99	Nylon Filament (500g)	https://ooznest.co.uk/product/nylon-3d-printer-filament-1-75mm/?attribute_pa_filament-colour=midnight-black&attribute_pa_filament-weight=500g-spool	Filament used in heat-resistant parts of MIST as we cannot mount motors directly to PLA or it will melt.
	1	7.1	7.1	White Lithium Grease	https://www.aliexpress.us/item/3256807003139498.html?	Lube for contact points. Use on spinning axles and metal to metal contact against bearings.
	1	7	7	Two part epoxy	https://www.amazon.com/Gorilla-Epoxy-Minute-ounce-Syringe/dp/B001Z3C3AG/	This is used everywhere a bearing is located to retain them. In addition, this is also used to attach the pulleys to the shafts. Let the epoxy cure before budging applied items.
	1	5.8	5.8	Loctite 242 for screws	https://www.aliexpress.us/item/3256808928775256.html?	This is used everywhere a screw or nut is located to prevent loosening against vibrations. Let the loctite cure before budging the screw or nut.
	1	4.22	4.22	Tapping fluid	https://www.aliexpress.com/i/3256808042437319.html?	Fluid used to cool down tap and prevent snapping while tapping shaft holes.
HARDWARE	2	4.53	9.06	83T HTD 3M Belts 6mm Width (249mm)	https://www.aliexpress.com/i/3256808042437319.html?	Belt connecting hip servo to hips.
	4	3.73	14.92	60T HTD 3M Belts 6mm Width (180mm)	https://www.aliexpress.us/item/3256804572314863.html?	Belt connecting leg motor to wheels.
	4	3.51	14.04	55T HTD 3M Belts 6mm Width (165mm)	https://www.aliexpress.us/item/3256806967864158.html?	Belt connecting leg motor to wheels.
	1	1.05	1.05	Wires (10 awg)	https://www.freeportmarine.com/products/anc108010-100?	10 AWG wire for wiring high current portions of MIST.
	5	0.8	4	Wires (12 awg)	https://www.freeportmarine.com/products/anc106010-100?	12 AWG wire for wiring medium current portions of MIST.
	2	1.21	2.42	Wires (18 awg)	https://www.nassauelectrical.com/products/belden-34418-18-awg-1-conductor-16x30-stranding-tinned-copper-600v-lead-wire?	18 AWG wire for wiring low current portions of MIST.
	8	1.35	10.8	M4 heat set inserts	https://www.mcmaster.com/94459A150/	The exact model doesn't matter, just ensure the height exceeds no more than 4.7mm.
	1	12.69	12.69	4mm ID x 8mm OD x 250mm shaft	https://www.walmart.com/ip/3-Pack-250mm-6063-Aluminum-Round-Tube-4mm-ID-8mm-OD-Seamless-Aluminum-Tubing-for-DIY-Crafting-0-16x0-31x9-8IN/16864413682?wmlspartner=wlpa&selectedSellerId=571	You will need to be able to cut the shaft accurately into smaller sections.

	18	0.6	10.8	8mm ID x 16 OD x 5mm WD (f688z Flanged Bearings)	https://www.aliexpress.us/item/3256812035160016.html?	Flanged bearings used across the robot to house shafts.
	1	7.75	7.75	Bungee Cord	https://www.revrobotics.com/rev-41-1163/	Bungee cord keeping arm's lateral position in place.
	1	5.33	5.33	Velcro	https://www.aliexpress.us/item/3256808114655678.html?	Velcro used to strap down battery in place.
	1	5.24	5.24	Scotch Extreme Mounting Tape	https://www.amazon.com/Scotch-051141397252-Extremely-Mounting-414P-ST/dp/B018GKCI82/	Mounting tape used to tape down wires and screwless electronics.
	1	2.89	2.89	M2.5 washer	https://www.mcmaster.com/93475A196/	Washer used in retaining a shaft.
	12	0.3158333333	3.79	10-32 washers	https://www.mcmaster.com/92141A011/	Washers used in retaining shafts.
	1	1.48	1.48	Electrical Tape	https://www.amazon.com/Duck-282289-Electrical-4-Inch-Single/dp/B007JSGNVQ/	Electrical tape used in covering exposed wires.
	1	1.77	1.77	Shrink Tubing	https://www.aliexpress.us/item/3256807959988149.html?	Shrink tubing used in covering exposed wires.
	1	5.05	5.05	PTFE thrust bearing slider	https://www.mcmaster.com/2797T127/	Must be identical to linked one.
	5	5	25	Sets of M2, M2.5, M3, M4, 10-32 screws	https://www.aliexpress.us/item/2255799852960451.html?	(You can get a set of both rounded & socket head screws online or match the screws you buy by length, size, and amount to the CAD 1:1 by searching "screw" in the assembly and counting.)
	4	8.37	33.48	128-2Z Bearings	https://www.mcmaster.com/7804K115/	Smaller-sized OD bearings used in housing shafts in gripper portion of MIST's arm.
TOOLS	1	1	1	10-32 tap (for tapping shafts)	https://www.aliexpress.us/item/2255800936437301.html?	Tap used in tapping screw holes in the shaft type being used across MIST (4mm ID x 8mm OD)
	1	1	1	10-32 tap handle	https://www.aliexpress.us/item/3256807238346556.html?	Tap handle necessary to house tap being used.
	1	10.99	10.99	Soldering Iron & Solder Kit	https://www.amazon.com/Soldering-Digital-Welding-Portable-Electric/dp/B08R3515SF/	Soldering iron and solder necessary to connect electronics across MIST.
	1	6.77	6.77	Wrench Set	https://www.aliexpress.us/item/3256806596445021.html?	Wrench set necessary to connect mechanical portions across MIST.
	1	15.09	15.09	Allen Key Set	https://www.aliexpress.us/item/3256811924676788.html	Allen key set used to tighten and loosen screws in combination with the wrench set.
	1	20.41	20.41	Wire Stripper, Cutter, and Crimper tool	https://www.aliexpress.us/item/3256809687963497.html?	Multifunctional electrical tool used for ensuring proper wire connections.
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